

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination
Nov 2013

CL 207 Concrete Technology

Date: 11.11.2013 Monday

Time: 01:30 p.m. To 4:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is role of water for durability of concrete. 03
- (b) When concrete is termed as special concrete? State various types of its. 04
- Q - 2 (a) Explain the manufacturing process of Portland cement with neat sketch. 04
- (b) Write procedure of laboratory tests to measure strength of cement. 03
- (c) State the properties and use of Rapid Hardening Cement. 04
- (d) What is Fly Ash? Explain influence of fly ash in mass concrete with respect to cement. 03

OR

- Q - 2 (a) What is setting time? Explain the term initial and final setting time of Ordinary Portland Cement. 03
- (b) Describe the role of gypsum in the hydration process of cement. 04
- (c) State the properties and use of Sulphate resisting cement. 04
- (d) From where silica fumes is obtained? Explain its influence on properties of fresh concrete. 03
- Q - 3 (a) Explain role of air entraining admixture in brief. 03
- (b) What is difference between Mineral Admixture and Chemical admixtures? 04
- (c) Explain various mechanical properties of aggregate. 07

OR

- Q - 3 (a) What are admixtures? Explain the benefits of using admixture in concrete. 07
- (b) Explain alkali aggregate reaction in aggregate. State factors affecting alkali aggregate reaction. 07

SECTION - II

- Q - 4 (a) In mix proportioning, generally we use minimum quantity of water, why? 03
- (b) Find the total yield of concrete in liter for one bag of cement with the proportion of 1:1.5:2.5. Consider specific gravity 3.2, 2.7, 2.8 and 1 for cement, FA, CA and water respectively. Entrapped air 2%. 04
- Q - 5 (a) In detail, explain manufacturing process of concrete. 07
- (b) Explain Fiber reinforced concrete in terms of ingredients and properties at its fresh state. 03
- (c) State various methods of mix design. Write method of any one. 04
- OR
- Q - 5 (a) Write a note on workability of concrete including definition, factors affecting it and methods of its measurement. 07
- (b) Write short note on ready mix concrete. 03
- (c) Explain how moisture correction of moist aggregate is considered in water demand in mix design. 04
- Q - 6 (a) State various causes of permeability. Why durability and permeability correlated with each other? 07
- (b) Why concrete is tested on the 28th day after casting? 02
- (c) State factors affecting the workability of concrete. 02
- (d) State various precautionary measures during mass concreting. 03
- Q - 6 (a) State factors affecting strength of concrete. 07
- (b) How concrete is classified according to its strength? 02
- (c) State parameters which results in to segregation of concrete. 02
- (d) State disadvantages of fiber reinforced concrete. 03
